ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 2 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, examples and equivalent code.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 D ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 D ✓ 1.1.8 B ✓ 1.1.9 D ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 I ✓ 1.3.2 G ✓ 1.3.3 F ✓ 1.3.4 B ✓ 1.3.5 E ✓ 1.3.6 H ✓ 1.3.7 C ✓ 1.3.8 D ✓ 1.3.9 A ✓ 1.3.10 J ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 A manufacturing robot is usually fixed in one place and repeats tasks in a controlled work cell ✓; a mobile robot moves around and must sense and react to a changing environment ✓. Field robot, e.g. farm robot, mining robot, underwater or space rover ✓; service robot, e.g. robot vacuum, hospital delivery robot, lawn mower ✓ (4)

2.2 Perception: sensors (bump, cliff, infrared, camera) sense walls, furniture and dirt ✓; cognition: the processor decides where to go and plans a cleaning path ✓; actuation: motors drive the wheels and brushes ✓. (3)

2.3 Any FOUR named ✓ and explained ✓: IoT endpoints (sensors / devices) – collect data or act; edge nodes – process data close to the device; gateways and network interfaces – connect devices to the network / internet and translate protocols; internet and IoT cloud – stores and processes data; applications – dashboards / apps that show data and let users control devices (8)

2.4 Any TWO ✓✓: smart meters / prepaid electricity monitoring; vehicle tracking; smart farming (soil moisture, irrigation); home security cameras and alarms on phones; health monitoring; smart factories (2)

[17]

QUESTION 3

3.1 (a) FDM ✓ – cheap, common printers, plastic filament ✓ (b) SLA ✓ – very fine detail and smooth surface from resin ✓ (c) DMLS (direct metal laser sintering) ✓ – produces strong, dense metal parts ✓ (6)

3.2 Any SIX steps in order ✓✓✓✓✓✓: input modelling (create the 3D CAD model / STL); orientation and positioning on the bed; adding support structures; slicing into layers; path planning (nozzle / laser paths, infill, perimeters); machine instructions (G-code) sent to the printer (6)

3.3 Any TWO ✓✓: slow printing; limited build size; warping; weaker layer bonding; expensive materials / machines; finishing needed. Reduction for ONE, e.g. warping – heated bed / enclosure / brim ✓ (3)

[15]

QUESTION 4

4.1.1 $\mathrm{V} = 4 \times 3{,}7 = 14{,}8\ \mathrm{V}$ ✓✓; $\text{capacity stays } 2\,500\ \mathrm{mAh} = 2{,}5\ \mathrm{Ah}$ ✓ (3)

4.1.2 $\mathrm{V} = 2 \times 3{,}7 = 7{,}4\ \mathrm{V}$ ✓✓; $\text{capacity} = 2 \times 2{,}5 = 5\ \mathrm{Ah}$ ✓ (3)

4.1.3 $\mathrm{t} = \mathrm{Ah} \div \mathrm{I}$ ✓ $= 5 \div 1{,}25 = 4\ \mathrm{h}$ ✓ (2)

4.2 DC flows in one direction only (e.g. batteries) ✓; AC changes direction many times per second (e.g. 50 Hz mains) ✓; single-phase AC has one alternating voltage (two wires: live and neutral), used in homes ✓; three-phase AC has three voltages 120° apart, used for large motors and industry ✓ (4)

4.3 $1/\mathrm{R_T} = 1/10 + 1/20 + 1/30$ ✓ $= 0{,}1833$ ✓; $\mathrm{R_T} = 5{,}45\ \Omega$ ✓; $\mathrm{I} = \mathrm{V} \div \mathrm{R}$ ✓ $= 12 \div 5{,}45 = 2{,}2\ \mathrm{A}$ ✓ (5)

[17]

QUESTION 5

5.1 A desktop PC is too big, heavy and power-hungry for a battery robot and has no direct input/output pins for sensors and motors ✓. Criteria (any FOUR) ✓✓✓✓: number of input/output pins (digital, analogue, PWM); processing speed and memory; power consumption and voltage (3,3 V / 5 V); communication (I2C, SPI, UART, Wi-Fi); size; cost and availability; programming support / libraries (5)

5.2.1 When the light reading drops below 300 (dark) ✓ the LED switches on; otherwise it is off ✓. (2)

5.2.2 It sends the light reading to the Serial Monitor so you can see / test the values ✓. (1)

5.2.3 Read sensor ✓, map / invert the value ✓, analogWrite on the PWM pin ✓, delay and correct syntax ✓, e.g. (4)

```
void loop() {
  int light = analogRead(sensorPin);
  int brightness = map(light, 0, 1023, 255, 0);
  analogWrite(ledPin, brightness);
  delay(200);
}
```

5.3 Components (any THREE) ✓✓✓: Arduino, ultrasonic sensor (HC-SR04), 16×2 I2C LCD, buzzer, breadboard / jumper wires / power supply. Steps ✓✓✓: send a trigger pulse and measure the echo time; calculate distance = time × 0,0343 ÷ 2 and show it on the LCD; if the distance < 20 cm switch the buzzer on, else off; repeat (6)

5.4 Seven-segment: shows only digits / a few letters, simple and very readable from far ✓; LCD 16×2: shows text and numbers on two lines (e.g. "Distance: 35 cm"), more information ✓; LCD with I2C needs fewer pins, seven-segment needs many pins or a driver ✓ (3)

[21]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE